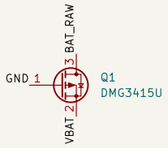
Button-activated Soil Moisture Meter  –  2xAAA (2.0–3.2V), no regulator
Idle: MCU in power-down, TPS22917 off –> ~0.5 uA total.  Press SW1 –> PA6 wakes MCU –>
LOAD_EN high –> VSW powers OLED + sensor –> read, display 15 s –> LOAD_EN low –> sleep.





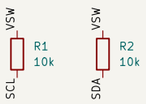


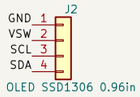


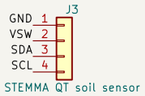


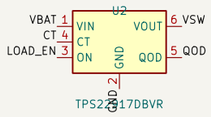


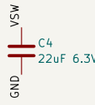


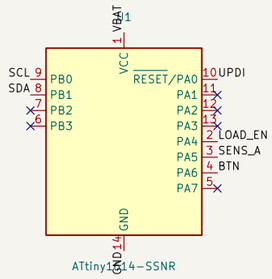











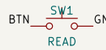











Firmware notes: run at 5 MHz (valid down to 1.8 V). Before LOAD_EN low, disable TWI and set
PB0/PB1 (and PA5) to input w/o pull-up so the unpowered OLED/sensor aren't back-fed.
Battery voltage measured internally (ADC: VDD vs. internal reference) – no divider needed.
OLED module: bridge its onboard 662K LDO (VIN->3V3) so it runs to ~2.2 V. Low-batt cutoff ~2.2 V.

2xAAA, ATtiny1614, TPS22917 load switch, SSD1306 OLED, STEMMA soil sensor

Sheet: /
File: Moisture_Sensor.kicad_sch

**Title: Soil Moisture Meter**

| Size: A3 | Date: 2026-10-26 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 1/1 |